Modern Regression

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Introduction

Goal

- Next topic: Modern Regression and Classification Methods.
- Challenge: Number of predictors is more than number of observations. (Large p, small n problem).
- Select the best subset of predictors.
- Regression - Basics $\rightarrow$ forward selection/backward selection/AIC $\rightarrow$ Ridge and LASSO (ℓ_1 penalized linear regression).

Formulation

- Multiple linear model assumes a linear relation between predictor and response.
- Simple linear model:

$$Y = \beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p + \epsilon.$$

- β_0 and β_j's are intercept and slopes, ϵ is a random variable with mean zero and independent from X.
- Another interpretation:

$$\mathbb{E}(Y|X_1 = x_1, \ldots, X_p = x_p) = \beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p,$$

since $\mathbb{E}(\epsilon|x) = 0$.

Conditional Expectation: Toy Example

- Consider regression of house **price** (Y) on **size** (X).
- We are interested in distribution of Y given X, not marginally.
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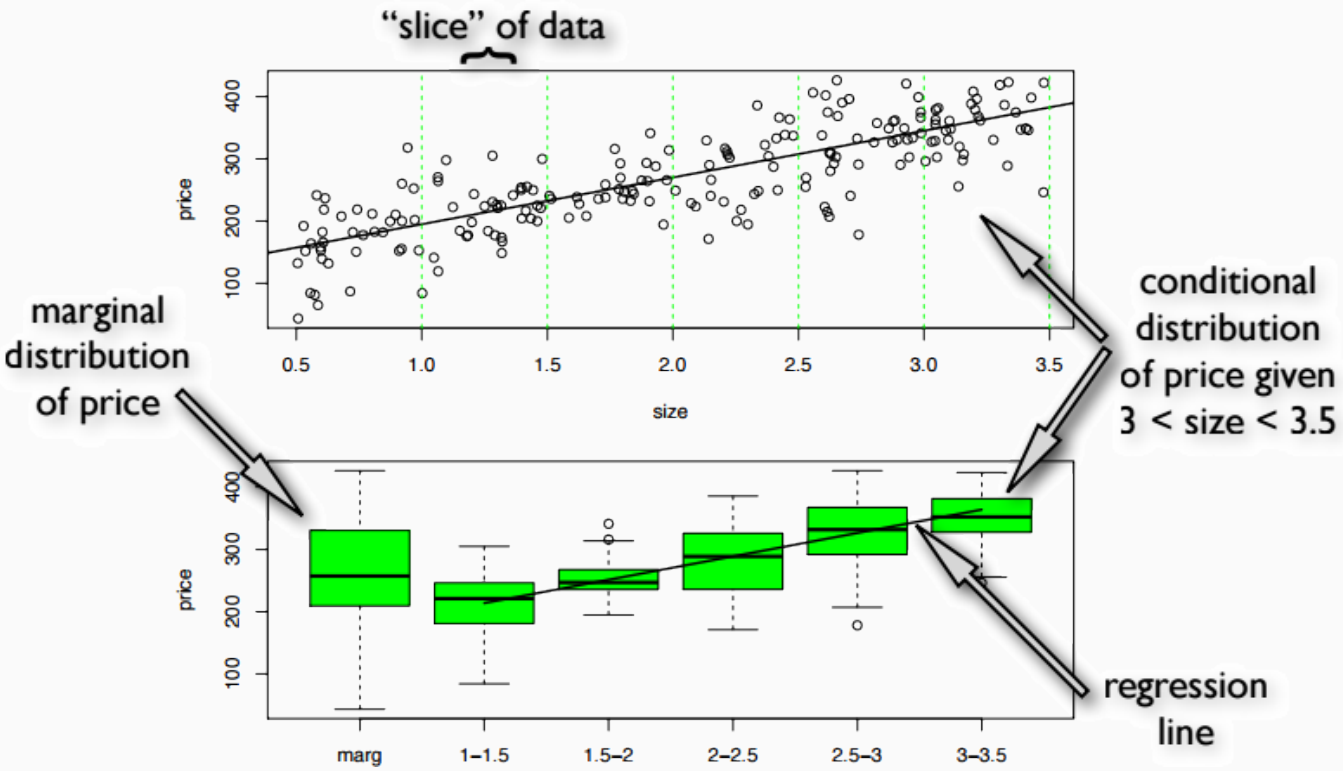
- Since X may help predict Y.
- **Conditional Distributions**
- Regression models the conditional distribution:

$$\mathbb{P}(Y \mid X).$$

- If we have observed data (x_i, y_i), $i = 1, \ldots, n$, we can learn about $\mathbb{P}(Y \mid X)$.

Conditional vs. Marginal Distribution

- A conditional distribution is obtained by slicing the X-Y point cloud.
- The **marginal** distribution ignores the slices.



Forecasting Using Conditional Distributions

- If I know a house is between 1-1.5k sq.ft., the conditional distribution provides:
  - A point forecast (median)
  - A prediction interval
- Conditional means align along a regression line.
- Conditional variance is smaller than marginal variance:

$$\mathbb{V}\mathrm{ar}\{Y \mid X\} \ll \mathbb{V}\mathrm{ar}\{Y\}.$$

Predictive Power of X

- If X has no predictive power:

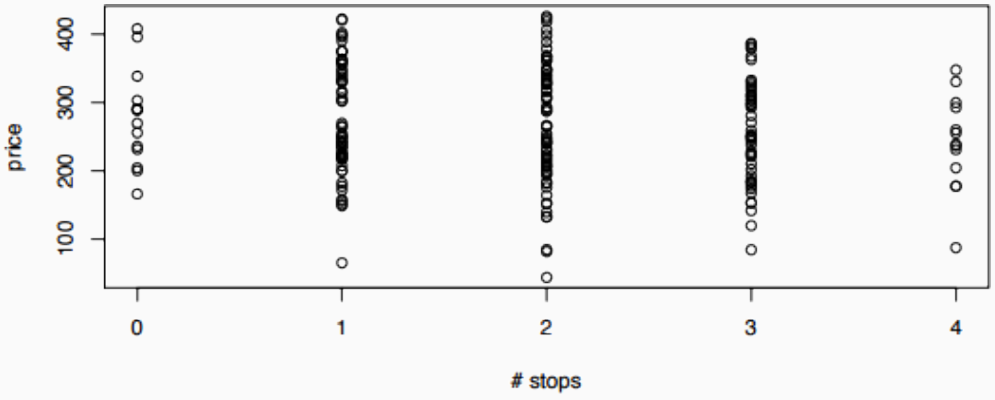
$$\mathbb{V}\mathrm{ar}\{Y \mid X\} \approx \mathbb{V}\mathrm{ar}\{Y\}.$$

- If X has forecasting information:

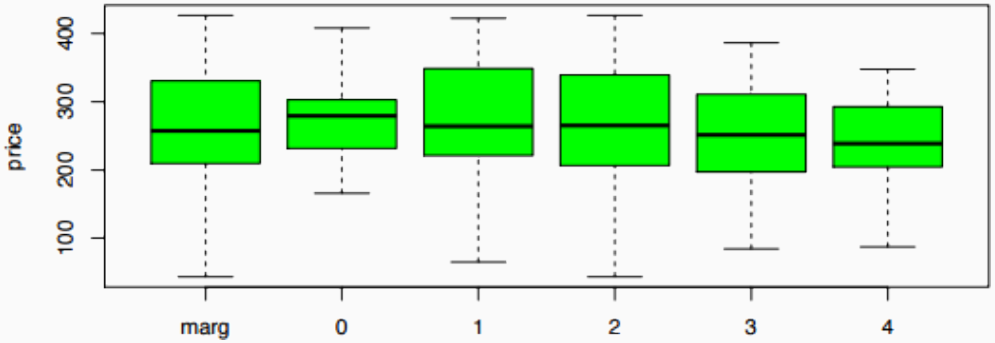
$$\mathbb{E}\{Y \mid X\} \neq \mathbb{E}\{Y\}, \quad \mathbb{V}\mathrm{ar}\{Y \mid X\} \ll \mathbb{V}\mathrm{ar}\{Y\}.$$

Example: X Has No Predictive Power

House price v. number of stop signs (Y) within a two-block radius of a house (X)



See that in this case the marginal and conditionals are not all that different



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- **From Simple to Multiple Linear Regression**
- Many problems involve multiple independent variables.
- Examples:
  - Multi-factor asset pricing models
  - Demand prediction based on prices, ads, demographics
  - More predictors for house price

The Multiple Linear Regression (MLR) Model

$$Y_i = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2),$$
$$Y_i \mid x_{i1}, \ldots, x_{ip} \sim \mathcal{N}(\beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p, \sigma^2).$$

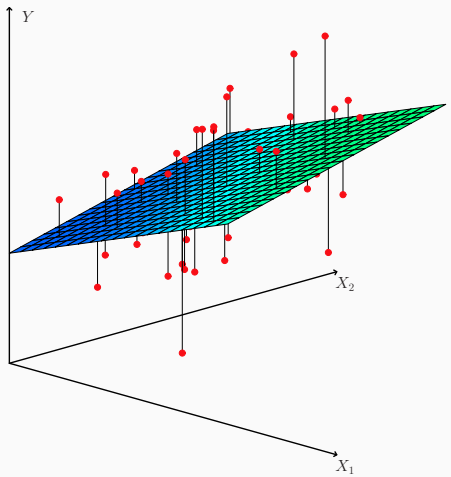
- Or in vectorized form:

$$Y \sim \mathcal{N}_n(X\beta, \mathbb{I}_n \sigma^2),$$

- where
  - $Y = (Y_1, \ldots, Y_n)^\top$
  - β is $(p+1) \times 1$-vector $(\beta_0, \beta_1, \ldots, \beta_p)^\top$
  - $\mathbb{I}_n$ is the $n \times n$ identity matrix
  - X is a $n \times (p+1)$ design matrix with first column of 1s.

Review: Multiple Linear Regression

- $Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \epsilon$, $\epsilon \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$.
- Equivalently, $\mathbb{E}(Y \mid X) = \mathbf{X}\boldsymbol{\beta}$.
- where $Y = (y_1, \ldots, y_n)^T$, $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_p)^T$ and X the usual $n \times p$ design matrix with an extra column of 1's to the left to account for intercepts.



Review: Multiple Linear Regression

- Is at least one of the variables X_i useful for predicting the outcome Y ?
- Which subset of the predictors is most important?
- How good is a linear model for these data?
- Given a set of predictor values, what is a likely value for Y, and how accurate is this prediction?

Estimate β

We find the estimates by minimizing the RSS [1]:

$$\begin{aligned}\text{RSS} &= \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \\ &= \sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \cdots - \hat{\beta}_p x_{i,p})^2\end{aligned}$$

One can show this is minimized by the vector $\hat{\boldsymbol{\beta}}$:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

[1] residual sum of squares

Estimate β

$$\hat{\beta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

We assume that the columns of the $\mathbf{X}_{n\times p}$ matrix $(X_1, \ldots, X_p)$ are linearly independent [2], so that $\text{rank}(\mathbf{X}) = p$. The fitted values are:

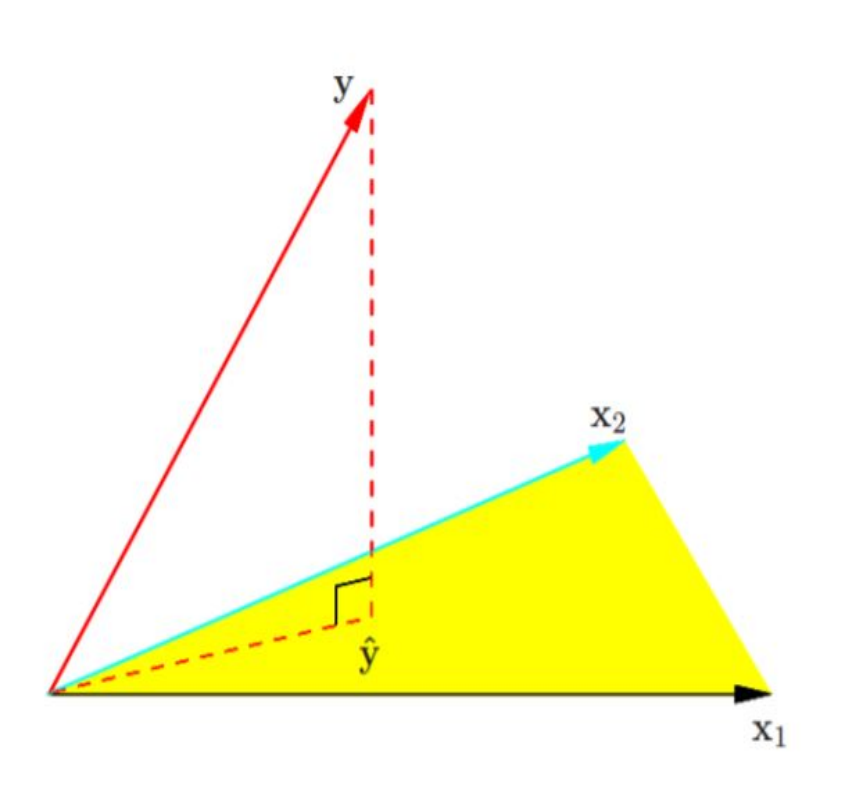
$$\hat{\mathbf{y}} = \mathbf{X}\hat{\beta} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

This is a linear function of $\mathbf{y}$, $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$, where $\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ is sometimes called the hat matrix.

[2] Note that this necessarily implies $p \leq n$

Geometry of Linear Regression

The fitted value $\hat{\mathbf{y}} \in \mathbb{R}^n$ in linear regression is exactly the projection of $\mathbf{y} \in \mathbb{R}^n$ on to the linear subspace spanned by the columns of $\mathbf{X}$, i.e. $\text{span}\{X_1, X_2, \ldots, X_p\} = \text{col}(X) \subseteq \mathbb{R}^n$.



The hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ is $P_{col}(X)$, the projection matrix onto the column space of $\mathbf{X}$.

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- Important: We need to invert $\mathbf{X}^T\mathbf{X}$, i.e. we need $p \leq n$!
- In other words, if $p > n$, there is not a unique least squares solution.

Which Variables are Important?

Consider the null hypothesis (q of the β_j's are zero)

$$H_0 : \beta_{p-q+1} = \beta_{p-q+2} = \cdots = \beta_p = 0$$

Let RSS_0 be the residual sum of squares for the model which **excludes these variables**. The F-statistic is defined by:

$$F = \frac{(RSS_0 - RSS)/q}{RSS/(n-p-1)}$$

Under the above null, this has a F distribution. We reject the null for a large value.

If $q = p$, we test whether any of the variables is important.

$$RSS_0 = \sum_{i=1}^{n} (y_i - \bar{y})^2$$

Special Case

- Special case: $q = 1$, exclude a single variable.
- Test whether this variable is related to Y after linearly correcting for all other variables.
- Equivalent to t-tests in R output.
- Must be careful with multiple testing.

Let us see an example!

Wine quality data

- `Year:` year in which grapes were harvested to make wine.
- `Price:` logarithm of the average market price for Bordeaux vintages according to 1990–1991 auctions. This is a nonlinear transformation of the response made to linearize the response.
- `WinterRain:` winter rainfall (in mm).
- `AGST:` Average Growing Season Temperature (in Celsius degrees).
- `HarvestRain:` harvest rainfall (in mm).
- `Age:` age of the wine measured as the number of years stored in a cask.
- `FrancePop:` population of France at Year (in thousands).

The quality of the wine is quantified as the Price, a clever way of quantifying a qualitative measure.

R output

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -2.343e+00  7.697e+00  -0.304  0.76384
Age          1.377e-02  5.821e-02   0.237  0.81531
AGST         6.144e-01  9.799e-02   6.270 3.22e-06 ***
FrancePop   -2.213e-05  1.268e-04  -0.175  0.86313
HarvestRain -3.837e-03  8.366e-04  -4.587  0.00016 ***
WinterRain   1.153e-03  4.991e-04   2.311  0.03109 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.293 on 21 degrees of freedom
Multiple R-squared:  0.8278,    Adjusted R-squared:  0.7868
F-statistic: 20.19 on 5 and 21 DF,  p-value: 2.232e-07
```

Figure 1: Linear Regression in R

How many variables are important?

- The t-statistic associated to the i-th predictor is the square root of the F-statistic for the null hypothesis which sets only $\beta_i = 0$.

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- We need to be careful about **multiple testing** issues.
- If P-value threshold is α, the model will declare $100\alpha\%$ of the variables as 'significant' just by pure chance $\Rightarrow$ false discoveries !
- Serious issue for large p.

Regression Setting

Theoretical Model

$$y = \beta_0 + \beta_1 X_1 + \cdots \beta_p X_p + \epsilon$$

$$y_i = \beta_0 + \sum_{j=1}^{p} \beta_j x_i + \epsilon_i,\ i = 1, \ldots, n$$

$$y_i = \sum_{j=0}^{p} \beta_j x_i + \epsilon_i,\ i = 1, \ldots, n$$

$$y = X\beta + \epsilon$$

$$y = f(X; \beta) + \epsilon$$

Fitted Model

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \cdots \hat{\beta}_p X_p$$

$$\hat{y}_i = \hat{\beta}_0 + \sum_{j=1}^{p} \hat{\beta}_j x_i,\ i = 1, \ldots, n$$

$$\hat{y}_i = \sum_{j=0}^{p} \hat{\beta}_j x_i,\ i = 1, \ldots, n$$

$$\hat{y} = X\hat{\beta}$$

$$\hat{y} = f(X; \hat{\beta})$$

Linear Regression:

- Lots of variants in how we write down a linear regression model
- Sometimes *p* is large, even *p* > *n*.
- Model selection: which x_js affect *y*?
- Errors: $\epsilon_i \overset{iid}{\sim} N(0, \sigma^2)$ or $E(\epsilon_i) = 0,\ \mathrm{Var}(\epsilon_i) = \sigma^2$
- σ^2 can be difficult to estimate
- Theory for nested models
- Use cross-validation, test data to select "best" models
- Many concepts extend to general (non-linear) regression.

In Praise of Linear Models

- Despite its simplicity, the linear model has distinct advantages in terms of its interpretability and often shows good predictive performance.
- Hence we discuss in this lecture some ways in which the simple linear model can be improved, by replacing ordinary least squares fitting with some alternative fitting procedures.

Why consider alternatives to least squares?

- Prediction Accuracy: especially when $p > n$, to control the variance.
- Model Interpretability: By removing irrelevant features — that is, by setting the corresponding coefficient estimates to zero — we can obtain a model that is more easily interpreted. We will present some approaches for automatically performing variable selection.

From Statistical Rethinking by R. McElreath

Rethinking: Stargazing. The most common form of model selection among practicing scientists is to search for a model in which every coefficient is statistically significant. Statisticians sometimes call this STARGAZING, as it is embodied by scanning for asterisks (**) trailing after estimates. A colleague of mine once called this approach the "Space Odyssey," in honor of A. C. Clarke's novel and film. The model that is full of stars, the thinking goes, is best.

But such a model is not best. Whatever you think about null hypothesis significance testing in general, using it to select among structurally different models is a mistake—p-values are not designed to help you navigate between underfitting and overfitting. As you'll see once you start using AIC and related measures, it is true that predictor variables that do improve prediction are not always statistically significant. It is also possible for variables that are statistically significant to do nothing useful for prediction. Since the conventional 5% threshold is purely conventional, we shouldn't expect it to optimize anything.

Modern Regression

Modern Regression

Idea: linear model can be improved, by replacing plain least squares fitting with some alternative fitting procedures!

Prediction Accuracy

- True relationship is approximately linear, LSE will have low bias. If $n >> p$, also low variance (perform well on test data).
- $n \approx p$: lot of variability in the least squares fit, resulting in overfitting and consequently poor predictions on test data.
- $p > n$, *no longer easy*: no unique LSE.
- $p > n$: the variance is infinite so the method cannot be used at all.
- Strategy: By constraining or shrinking the estimated coefficients, we can often substantially reduce the variance at the cost of a negligible increase in bias.

Model Interpretability

- Some or many of the variables used in a multiple regression model are in fact not associated with the response.
- Including such irrelevant variables leads to unnecessary complexity.
- By removing these variables, we can obtain a model that is more easily interpreted.
- LSE extremely unlikely to yield any coefficient estimates that are exactly zero.
- We'll see some approaches for automatically performing feature selection or variable selection!

Approaches

Three Main Approaches for Improving Linear Model:

1. Subset Selection:
2. Shrinkage:
3. Dimension Reduction:

Approaches

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- Shrinkage: fitting a model involving all p predictors. However, the estimated coefficients are shrunken towards zero. Shrinkage (also known as regularization) has the effect of reducing variance. Depending on what type of shrinkage is performed, some of the coefficients may be estimated to be exactly zero (can also perform variable selection).

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- Dimension Reduction: Projecting the p predictors into a M-dimensional subspace, where $M < p$. Then these M projections are used as predictors to fit a linear regression model by least squares.

How many?

When choosing a subset of the predictors, we have 2^p choices. We cannot test every possible subset!

Instead we can use a stepwise approach:

1. Construct a sequence of p models with increasing or decreasing number of variables.
2. Select the best model among them.

Variants of Stepwise Selection

- **Forward selection:** Starting from a null model (the intercept), include variables one at a time, minimizing the RSS at each step.
- **Backward selection:** Starting from the full model, eliminate variables one at a time, choosing the one with the largest t-test p-value at each step.
- **Mixed selection:** Starting from a null model, include variables one at a time, minimizing the RSS at each step. If the p-value for some variable goes beyond a threshold, eliminate that variable.

How many are important?

The output of a stepwise selection method is a range of models:

- $\{\}$.
- $\{tv\}$
- $\{tv, newspaper\}$.
- $\{tv, newspaper, radio\}$
- $\{tv, newspaper, radio, facebook\}$
- $\{tv, newspaper, radio, facebook, twitter\}$

6 choices are better than $2^6 = 64$. We use different tuning methods to decide which model to use; e.g. **cross-validation**, AIC, BIC.

Model Selection

- **Cross-validation** is an important tool for all statistical learning methods - detailed discussion later.
- We will also discuss AIC, BIC and Stepwise in some details but I wanted to discuss a couple of issues.

Dealing with Categorical Predictors

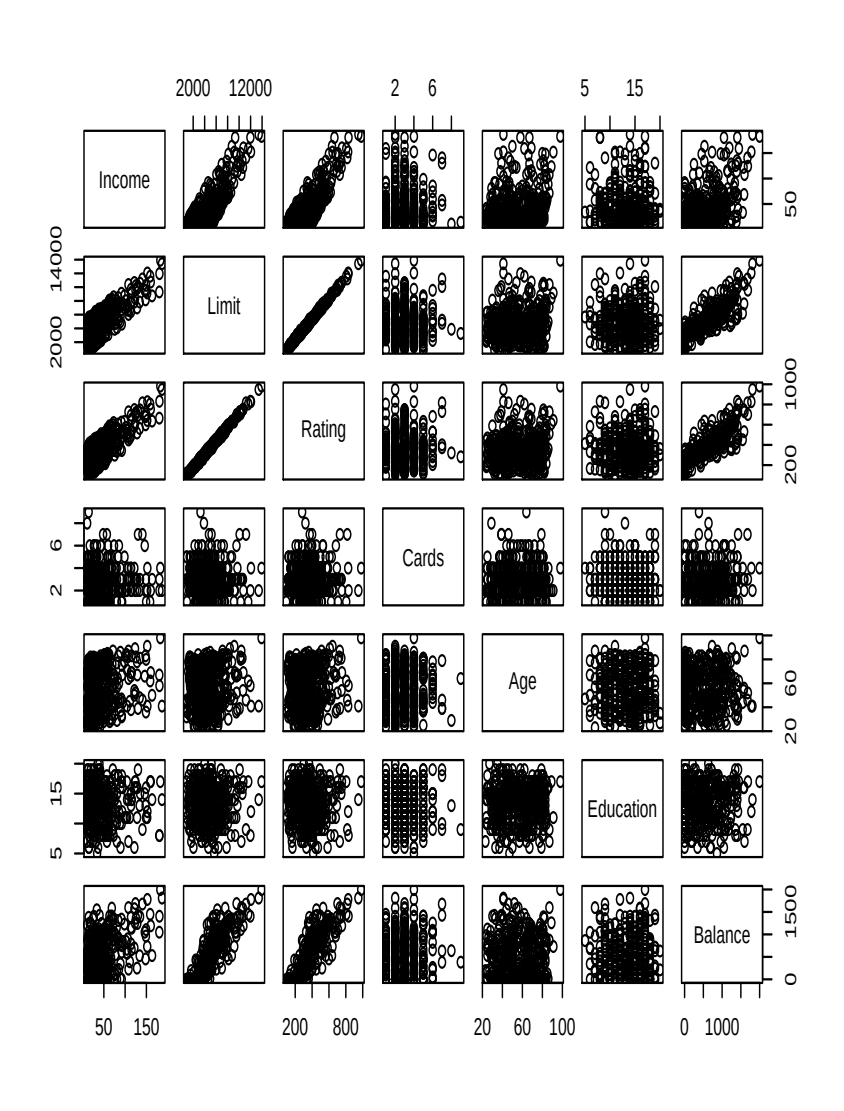

Figure 2: Credit Data

In addition, there are 4 qualitative variables:

1. gender: male, female.
2. student: student or not.
3. status: married, single, divorced.
4. ethnicity: African-American, Asian, Caucasian.

Dealing with Qualitative Variables

For each qualitative predictor, e.g. status:

- Choose a baseline category, e.g. single
- For every other category, define a new predictor:
- $X_{married}$ is 1 if the person is married and 0 otherwise.
- $X_{divorced}$ is 1 if the person is divorced and 0 otherwise. The model will be:

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_7 X_7 + \beta_{married} X_{married} + \beta_{divorced} X_{divorced} + \epsilon \quad (1)$$

- $\beta_{married}$ is the relative effect on balance for being married compared to the baseline category.

Dealing with Qualitative Variables

- The model fit and predictions are independent of the choice of the baseline category.
- However, the interpretation of parameters and associated hypothesis tests depend on the baseline category.
- Solution: To check whether status is important, use an F-test for the hypothesis $H_0 : \beta_{married} = \beta_{divorced} = 0$. This does not depend on the coding of the baseline category.

How uncertain are the predictions

The function `predict` in R output predictions from a linear model:

```
> predict(lm.fit,data.frame(lstat=(c(5,10,15))), interval="confidence")
       fit      lwr      upr
1 29.80359 29.00741 30.59978
2 25.05335 24.47413 25.63256
3 20.30310 19.73159 20.87461
> predict(lm.fit,data.frame(lstat=(c(5,10,15))), interval="prediction")
       fit       lwr      upr
1 29.80359 17.565675 42.04151
2 25.05335 12.827626 37.27907
3 20.30310  8.077742 32.52846
```

1. "Confidence intervals" reflect the uncertainty on $\hat{\beta}$; i.e. confidence interval for $f(x_0)$.
2. "Prediction intervals" reflect uncertainty on $\hat{\beta}$ and the irreducible error ϵ as well; i.e. confidence interval for y_0.

The Story So Far

- Defined Multiple Linear Regression
- Discussed how to test the relevance of variables.
- Described one approach to choose a subset of variables.
- Explained how to code qualitative variables.
- Discussed confidence intervals surrounding predictions.
- Now, how do we evaluate model fit? Is the linear model any good? What can go wrong?

How good is the fit?

To assess the fit, we focus on the residuals.

- $R^2 = \text{Corr}(Y, \hat{Y})$ always increases as we add more variables.
- The Residual Standard Error doesn't always improve with more predictors:

$$RSE = \sqrt{\frac{1}{n-p-1}\text{RSS}}$$

- One way to investigate departure from model assumptions is to **visualize the residuals**.

Potential Issues in Linear Regression

1. Potential issues in linear regression
2. Interactions between predictors
3. Non-linear relationships
4. Correlation of error terms
5. Non-constant variance of error (heteroskedasticity).
6. Outliers
7. High leverage points
8. Collinearity !!

What do we know so far

- In linear regression, adding predictors always decreases the training error or RSS.
- However, adding predictors does not necessarily improve the test error.
- When $n < p$, there is no least squares solution.

$$\hat{\beta} = \underbrace{(\mathbf{X}^T\mathbf{X})^{-1}}_{\text{singular}}\mathbf{X}^T\mathbf{y}$$

- So, we must find a way to select fewer predictors if we want to use least squares linear regression

Next on Deck:

1. Best Subset Selection.
2. Forward Selection.
3. Backward Selection.
4. Shrinkage Methods: Ridge Regression and LASSO.

Best Subset Selection

- Simple idea: let's compare all models with k predictors.
- There are $\binom{p}{k} = p!/[k!(p-k)!]$ possible models.
- Choose the model with the smallest RSS. Do this for every possible k.

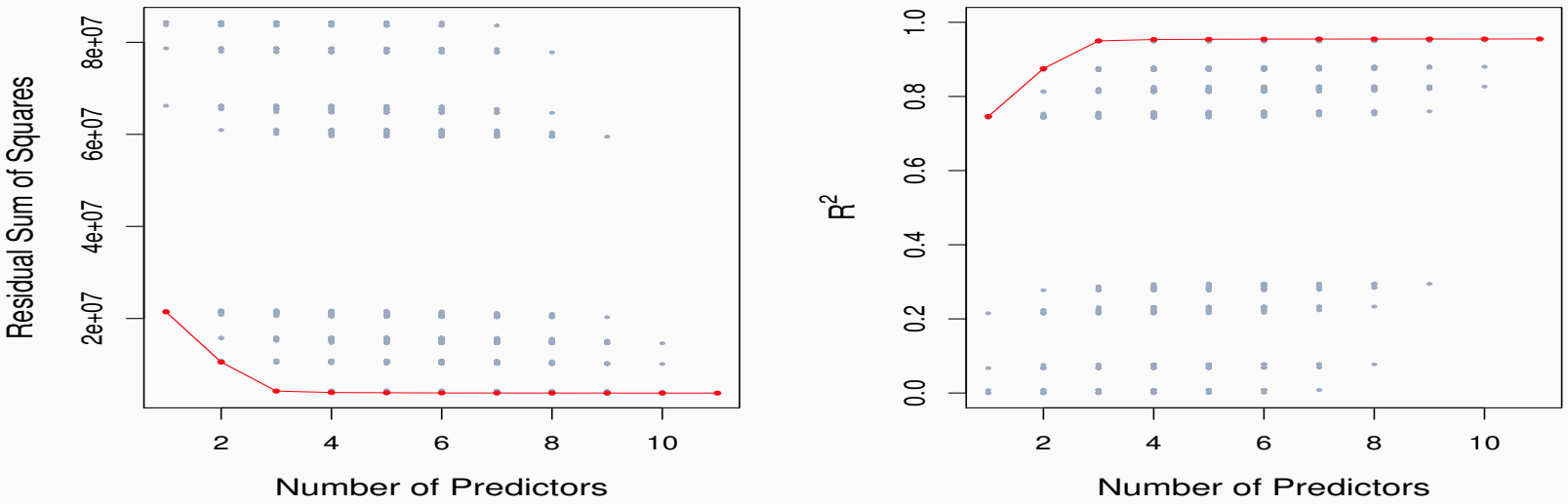

Figure 3: For each possible model containing a subset of the ten predictors in the Credit data set, the RSS and R^2 are displayed. The red frontier tracks the best model for a given number of predictors, according to RSS and R^2.

Best Subset Selection i

To optimize k, we want to minimize the test error, not the training error.

We could use **cross-validation**, or alternative estimates of test error:

1. Akaike Information Criteria (AIC):

$$AIC = \frac{1}{n\hat{\sigma}^2}(RSS + 2k\hat{\sigma}^2)$$

   or, Mallow's C_p:

$$C_p = \frac{1}{n}(RSS + 2k\hat{\sigma}^2)$$

   where $\hat{\sigma}^2$ is an estimate of the irreducible error, typically $\hat{\sigma}^2 = (n - p - 1)^{-1} \times$ RSS of full model

Best Subset Selection ii

2. Bayesian Information Criterion (BIC):

$$BIC = \frac{1}{n}(RSS + \log(n)k\hat{\sigma}^2)$$

3. Adjusted R^2:

$$R^2 = 1 - \frac{RSS}{TSS}$$

where $TSS = \sum_i (y_i - \bar{y})^2$.

$$R^2_{adj} = 1 - \frac{RSS/(n-k-1)}{TSS/(n-1)}$$

Application: Credit Data

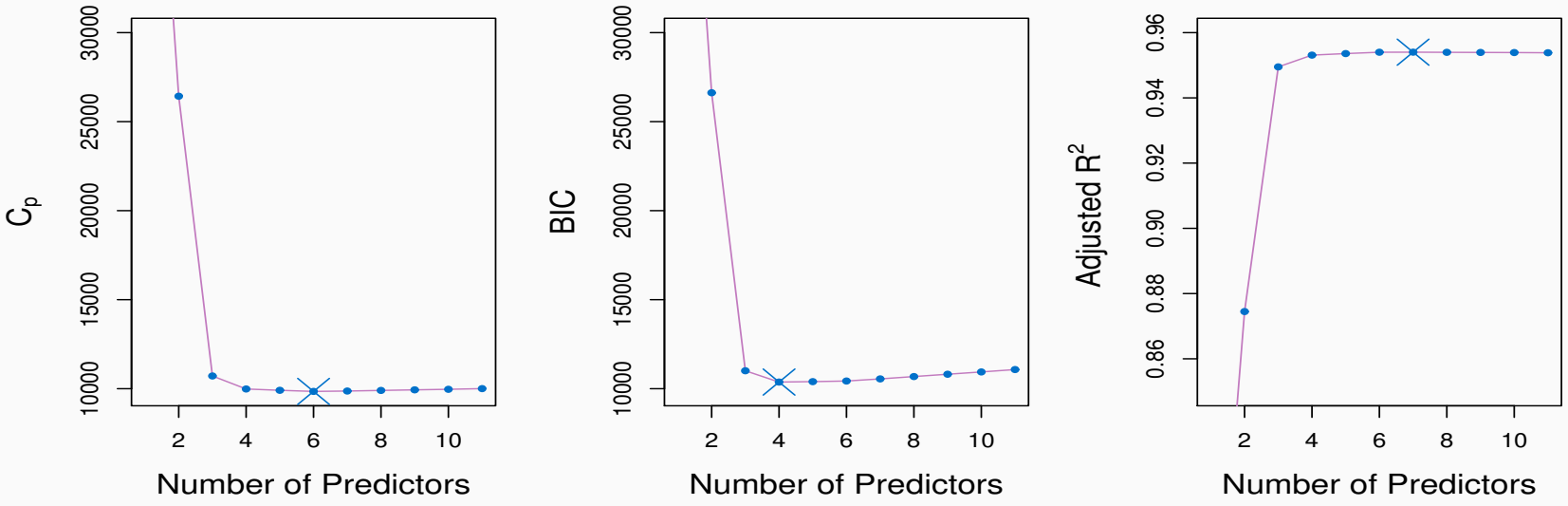

Figure 4: C_p, BIC, and adjusted R^2 are shown for the best models of each size for the `Credit` data set (the lower frontier in Figure 6.1). C_p and BIC are estimates of test MSE. In the middle plot we see that the BIC estimate of test error shows an increase after four variables are selected. The other two plots are rather flat after four variables are included.

Recap: Best Subset Selection

Algorithm 6.1 *Best subset selection*

1. Let $\mathcal{M}_0$ denote the *null model*, which contains no predictors. This model simply predicts the sample mean for each observation.

2. For $k = 1, 2, \ldots p$:

   (a) Fit all $\binom{p}{k}$ models that contain exactly k predictors.

   (b) Pick the best among these $\binom{p}{k}$ models, and call it $\mathcal{M}_k$. Here *best* is defined as having the smallest RSS, or equivalently largest R^2.

3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2.

Stepwise selection methods

Best subset selection has 2 problems:

1. It is often very expensive computationally. We have to fit 2^p models!
2. If for a fixed k, there are too many possibilities, we increase our chances of over-fitting (just like in multiple testing!). The model selected has high variance.
3. Solution: we can restrict our search space for the best model.
4. Reduces the variance of the selected model at the expense of an increase in bias.

Forward Selection

Algorithm 6.2 *Forward stepwise selection*

1. Let $\mathcal{M}_0$ denote the *null* model, which contains no predictors.
2. For $k = 0, \ldots, p-1$:
   (a) Consider all $p-k$ models that augment the predictors in $\mathcal{M}_k$ with one additional predictor.
   (b) Choose the *best* among these $p-k$ models, and call it $\mathcal{M}_{k+1}$. Here *best* is defined as having smallest RSS or highest R^2.
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2.

Advantages

- The Best Subset Selection procedure involves fitting 2^p models.
- Forward Stepwise Selection involves: 1 null model plus $p - k$ models in the kth iteration, $k = 0, \ldots, p - 1$.
- Total: $1 + \sum_{k=0}^{p-1}(p - k) = 1 + p(p + 1)/2$.
- For $p = 20$, best subset selection requires fitting 1,048,576 models, whereas forward stepwise selection requires fitting only 211 models.

Selection Criteria

- In Step 2(b) of Forward Stepwise Algorithm, we must identify the best model from among those $\binom{p}{k}$ that augment $\mathbf{M}_k$ with one additional predictor.

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Selection Criteria

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- We can do this by simply choosing the model with the lowest RSS or the highest R^2. (Fixed dimension)
- However, in Step 3, we must identify the best model among a set of models with different numbers of variables. This is more challenging – need AIC, BIC etc.

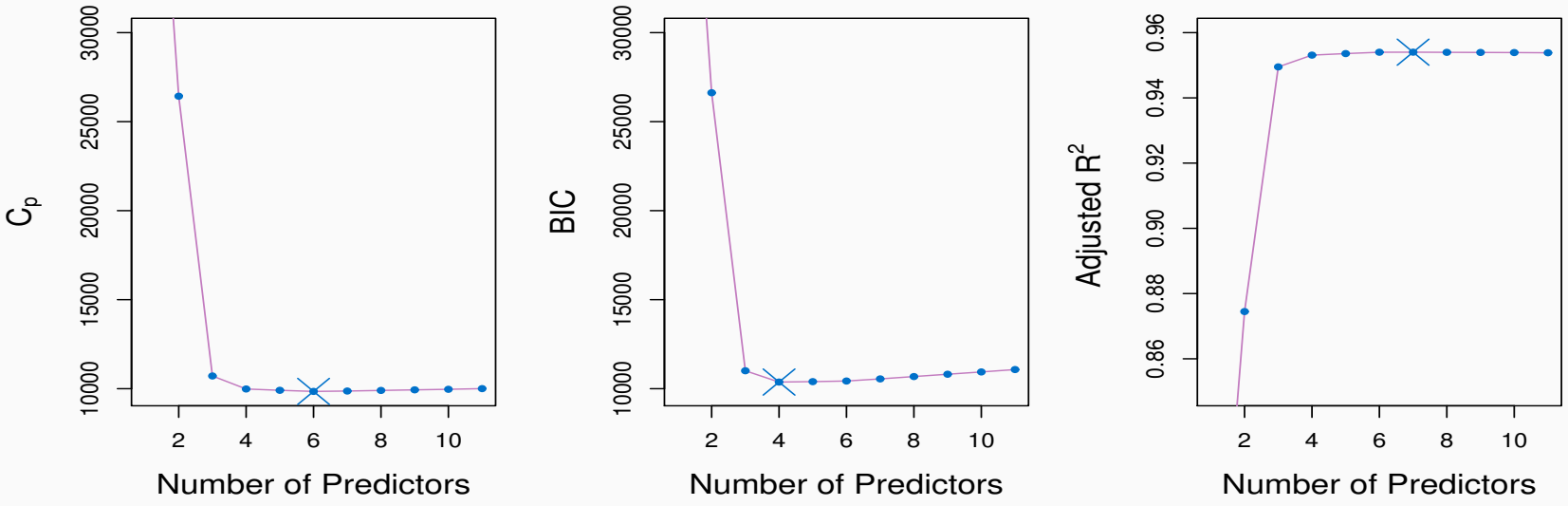
AIC, BIC etc.

Method	Expression	Comments
Akaike Information Criteria (AIC)	$1/(n\hat{\sigma}^2)(RSS + 2k\hat{\sigma}^2)$	Prediction
(Mallow's) C_p	$1/n(RSS + 2k\hat{\sigma}^2)$	Test error
Bayesian Information Criterion (BIC)	$1/n(RSS + \log(n)k\hat{\sigma}^2)$	**Heavier penalty**
Adjusted R^2	$1 - \frac{RSS/(n-k-1)}{TSS/(n-1)}$	

Table 1: Popular Selection Criteria

where $\hat{\sigma}^2 = (n - p - 1)^{-1} \times$ RSS of full model and $R^2 = 1 - \frac{RSS}{TSS}$ where $TSS = \sum_i (y_i - \bar{y})^2$.

Back to the Credit Data



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Which one to use?

- C_p, AIC, and BIC all have rigorous theoretical justifications!
- These justifications rely on asymptotic arguments (scenarios where the sample size n is very large).
- Despite its popularity, and even though it is quite intuitive, the adjusted R^2 is not as well motivated in statistical theory as AIC, BIC, and C_p.
- We have presented the formulas in the case of a linear model fit using least squares, however, these quantities can also be defined for more general types of models (e.g. logistic).

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- We have presented the formulas in the case of a linear model fit using least squares, however, these quantities can also be defined for more general types of models (e.g. logistic).
- My favourite: BIC.

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- For instance, suppose that in a given data set with $p = 3$ predictors, the best possible one-variable model contains X_1, and the best possible two-variable model instead contains X_2, X_3.
- Forward selection will fail to select the best possible two-variable model, because M_1 will contain X_1, so M_2 must also contain X_1 together with one additional variable.

# Variables	Best subset	Forward stepwise
One	rating	rating
Two	rating, income	rating, income
Three	rating, income, student	rating, income, student
Four	cards, income, student, limit	rating, income, student, limit

Figure 6: The first four selected models for best subset selection and forward stepwise selection on the `Credit` data set. The first three models are identical but the fourth models differ.

Backward Selection

Algorithm 6.3 *Backward stepwise selection*

1. Let $\mathcal{M}_p$ denote the *full* model, which contains all p predictors.
2. For $k = p, p-1, \ldots, 1$:
   (a) Consider all k models that contain all but one of the predictors in $\mathcal{M}_k$, for a total of $k-1$ predictors.
   (b) Choose the *best* among these k models, and call it $\mathcal{M}_{k-1}$. Here *best* is defined as having smallest RSS or highest R^2.
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2.

Properties

- Computational advantages of Forward Selection holds true for Backward Selection.
- Number of models evaluated $1 + p(p+1)/2 = O(p^2)$.
- Once out, stays out: Not guaranteed to yield the best model containing a subset of the p predictors.

‘High-dimensional’ data

- Forward stepwise selection can be applied even in the high-dimensional setting where $n < p$.
- However, in this case, we can only consider submodels $M_0, ..., M_{n-1}$ only, since each submodel is fit using least squares, which will not yield a unique solution if $p \geq n$.

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- However, in this case, we can only consider submodels $M_0, ..., M_{n-1}$ only, since each submodel is fit using least squares, which will not yield a unique solution if $p \geq n$.
- Backward selection requires that the number of samples n is larger than the number of variables p (so that the full model can be fit).
- In contrast, forward stepwise can be used even when $n < p$, and so is the **only viable subset method** when p is very large.

Forward vs. Backward

- You cannot apply backward selection when $p > n$.
- Although it seems like they should, they need not produce the same sequence of models.
- $X_1, X_2 \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma)$

$$X_3 = X_1 + 3X_2$$
$$Y = X_1 + 2X_2 + \epsilon$$

  Regress Y onto X_1, X_2, X_3
- Forward: $\{X_3\} \to \{X_3, X_2\} \to \{X_3, X_2, X_1\}$
- Backward: $\{X_1, X_2, X_3\} \to \{X_1, X_2\} \to \{X_3\}$.

Shrinkage Methods

The idea is to perform a linear regression, while regularizing or shrinking the coefficients $\hat{\beta}$ toward 0. Why would shrunk coefficients be better?

1. This introduces bias, but may significantly decrease the variance of the estimates. If the latter effect is larger, this would decrease the test error.
2. There are Bayesian motivations to do this: the prior tends to shrink the parameters.

Ridge Regression

- Ridge Regression solves the following optimization:

$$\min_{\beta} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{i,j} \right)^2 + \underbrace{\lambda \sum_{j=1}^{p} \beta_j^2}_{\text{shrinkage penalty}}$$

- The term in blue: RSS of the model.
- In red, the squared ℓ_2 norm of β or $||\beta||_2^2$.
- The parameter λ is a tuning parameter - it modulated the importance of fit versus shhrinkage.
- When $\lambda = 0$, the penalty term has no effect, and ridge regression will produce the least squares estimates.
- As $\lambda \to \infty$, the impact of the shrinkage penalty grows, and the ridge regression coefficient estimates will approach zero.

The Best Model

- Unlike least squares, which generates only one set of coefficient estimates, ridge regression will produce a different set of coefficient estimates.

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The Best Model

- Unlike least squares, which generates only one set of coefficient estimates, ridge regression will produce a different set of coefficient estimates.
- We find an estimate $\hat{\beta}_{\lambda}^{R}$ for many values of λ and then choose an optimal value of λ that minimises the test error.
- Fortunately, this is no more expensive than running a least-squares regression.

The intercept

- Look at the Ridge Regression optimization once again:

$$\min_{\beta} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{i,j} \right)^2 + \underbrace{\lambda \sum_{j=1}^{p} \beta_j^2}_{\text{shrinkage penalty}}$$

- The shrinkage penalty is applied to $\beta_1, \ldots, \beta_p$, but not to the intercept β_0.
- We want to shrink the estimated association of each variable with the response; however, we do not want to shrink the intercept.
- Intercept is simply a measure of the mean value of the response when $x_{i1} = x_{i2} = \ldots = x_{ip} = 0$.

Scaling

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- In ridge regression, this is not true.
- In practice, what do we do?
- Scale each variable such that it has sample variance 1 before running the regression.
- This prevents penalizing some coefficients more than others.

Solution Path

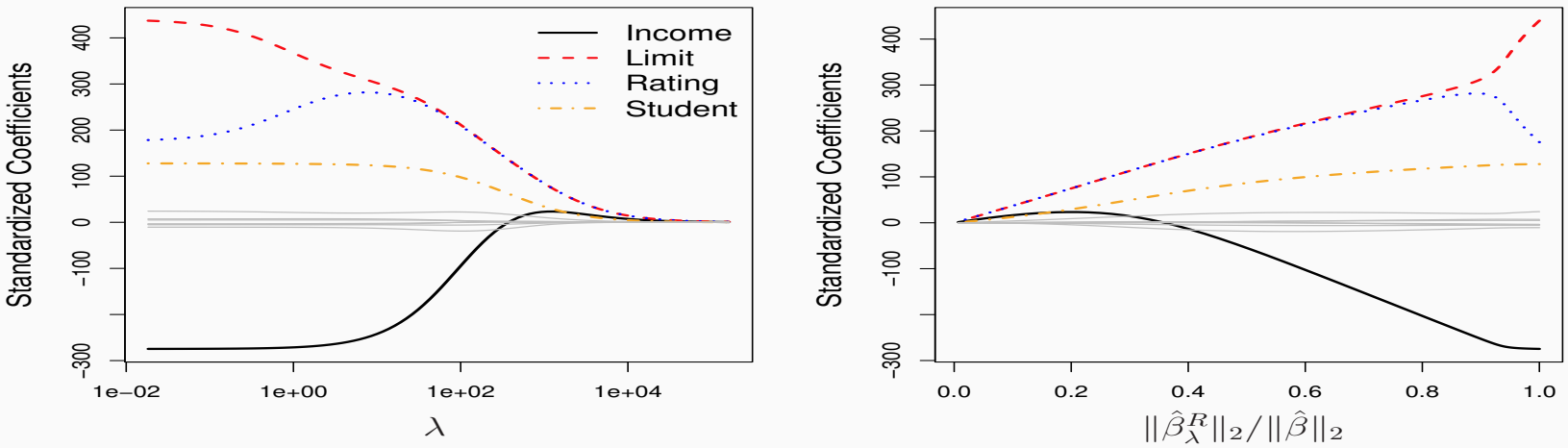

Figure 7: The standardized ridge regression coefficients are displayed for the `Credit` data set, as a function of λ and $\left\|\hat{\beta}^R_\lambda\right\|^2 / \left\|\hat{\beta}\right\|^2$.

Bias-Variance Trade-off

- Ridge regression's advantage over least squares is rooted in the bias-variance trade-off.
- As λ increases, the flexibility of the ridge regression fit decreases, leading to decreased variance but increased bias.

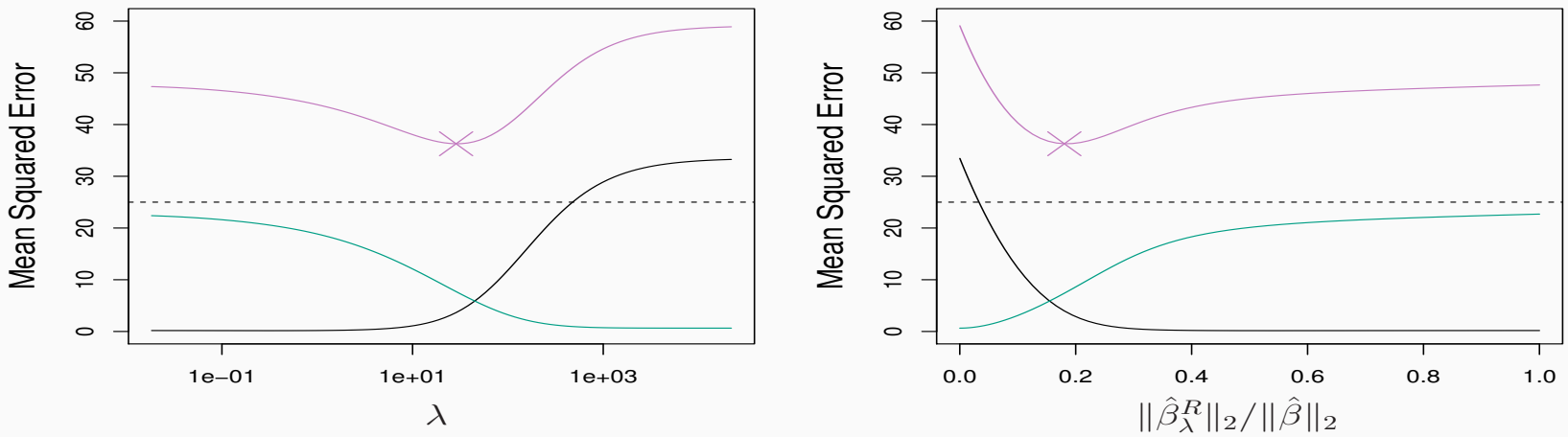

Figure 8: Squared bias, variance, and test mean squared error for the ridge regression predictions on a simulated data set, as a function of λ and $\left\|\hat{\beta}^R_\lambda\right\|^2 / \left\|\hat{\beta}\right\|^2$.

Choosing λ

- It is critical to choose a good value of λ - ideally we should pick λ that minimizes the test error.
- Recall distinction between training error and test error.
- If we do not have a large designated test data-set, **cross-validation** can be used to estimate the test error.
- **Cross-validation**: estimate the test error rate by holding out a subset of the training observations from the fitting process, and then applying the statistical learning method to those held out observations.

Lasso and Wide Data

Lasso

Wide Data: Hastie, Tibshirani

- Modern datasets are *wide* - i.e. many more features than samples - the linear model is a popular tool.
- **Document Classification:** bag-of-words model yields p = 20K features and N = 5K document samples.
- **Genomics, microarray studies:** p = 40K genes are measured for each of N = 100 subjects.
- **Genome-wide association studies:** p = 500K SNPs measured for N = 2000 case-control subjects.
- Different types of linear models are popular tools for such examples: e.g. linear regression, logistic regression, Cox model. Since $p >> N$, we cannot fit these models using standard approaches.

Many forms of regularization

We cannot fit linear models with $p > N$ without some constraints. Common approaches are:

- **Forward stepwise** adds variables one at a time and stops when overfitting is detected. This is a *greedy* algorithm, since the model with say 5 variables is not necessarily the best model of size 5.

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- **Best-subset** regression finds the subset of each size k that fits the model the best. Only feasible for small p around 35.
- **Ridge regression** fits the model subject to constraint $\sum_{j=1}^{p} \beta_j^2 \leq t$. Shrinks coefficients toward zero, and hence controls variance. Allows linear models with arbitrary size p to be fit, although coefficients always in row-space of X. In other words, coefficients are non-zero : Ridge doesn't perform variable selection.

The Lasso

- Lasso regression solves the following optimization :

$$\min_{\beta} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{i,j} \right)^2 + \underbrace{\lambda \sum_{j=1}^{p} |\beta_j|}_{\text{shrinkage penalty}}$$

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- Why use Lasso instead of Ridge?
- Ridge regression shrinks all the coefficients to a non-zero value.
- The Lasso shrinks some of the coefficients all the way to zero. Alternative to best subset selection or stepwise selection!

Ridge Regression

Ridge regression of `default` in the `Credit` dataset.

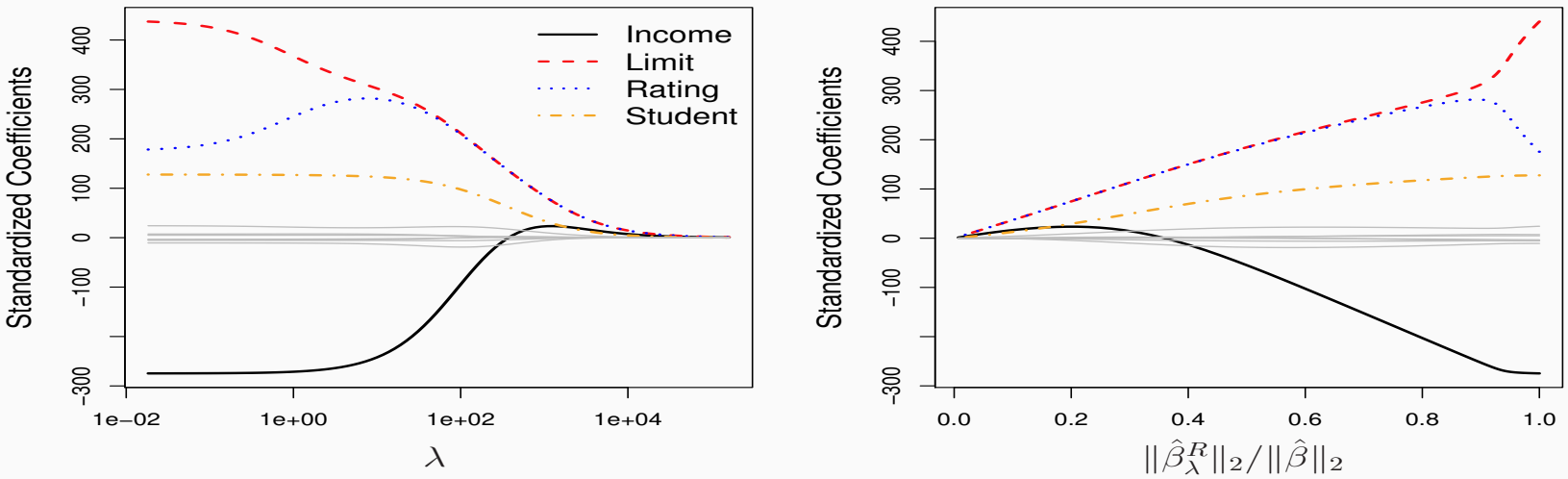

Figure 9: The standardized ridge regression coefficients are displayed for the `Credit` data set, as a function of λ and $\left|\left|\hat{\beta}^R_\lambda\right|\right|^2 / \left|\left|\hat{\beta}\right|\right|^2$.

A lot of pesky small coefficients throughout the regularization path.

Lasso Regression

Lasso regression of `default` in the `Credit` dataset.

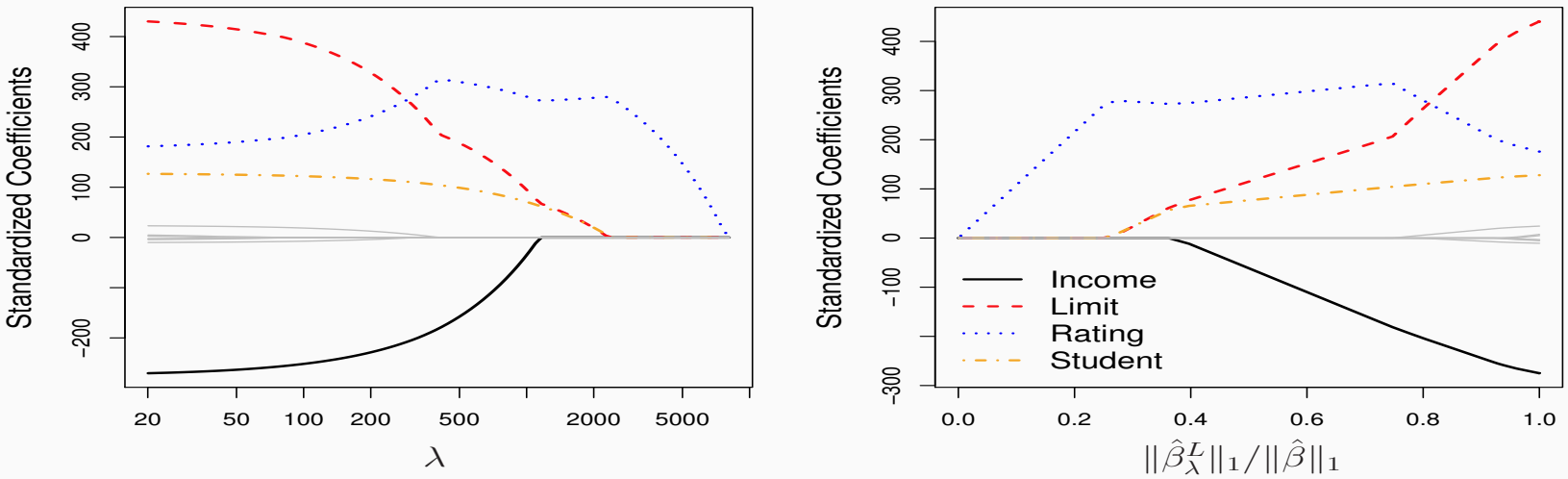

Figure 10: The standardized Lasso regression coefficients are displayed for the `Credit` data set, as a function of λ and $\left|\left|\hat{\beta}^R_\lambda\right|\right|^2 / \left|\left|\hat{\beta}\right|\right|^2$.

The small coefficients are shrunk to zero!

Lasso performs variable selection

- When $\lambda = 0$, Both the lasso and Ridge simply gives the least squares fit, and when $\lambda \to \infty$, both give the null model in which all coefficient estimates equal zero.
- However, in between these two extremes, the ridge regression and lasso models are quite different from each other.
- Left to right in last figure: at first the lasso results in a model that contains only the `rating` predictor. Then `student` and `limit` enter the model almost simultaneously, shortly followed by `income`.
- Eventually, the remaining variables enter the model.
- Hence, depending on the value of λ, the lasso can produce a model involving any number of variables.

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- Eventually, the remaining variables enter the model.
- Hence, depending on the value of λ, the lasso can produce a model involving any number of variables.
- In contrast, ridge regression will always include all of the variables in the model, although the magnitude of the coefficient estimates will depend on λ.

Sparse Models

- Hence, much like best subset selection, the lasso performs variable selection.
- As a result, models generated from the lasso are generally much easier to interpret than those produced by ridge regression.
- We say that the lasso yields sparse models – that is, sparse models that involve only a subset of the variables.
- As in ridge regression, selecting a good value of λ for the lasso is critical; and we shall use CV to do that.

Variable Selection

- Why is it that the lasso, unlike ridge regression, results in coefficient estimates that are exactly equal to zero?

Variable Selection

- Why is it that the lasso, unlike ridge regression, results in coefficient estimates that are exactly equal to zero?
- The alternative formulations can be used to shed light on the issue.

Sparsity

- But how does it select?
- It is helpful to make a comparison to the "L2 norm" version aka **ridge regression**.

$$\hat{\beta}_{\lambda}^{\text{ridge}} = \text{argmin}_{\beta} \, ||y - X\beta||^2 + \lambda \sum_j |\beta_j|^2$$

- Sometimes this is called "Tikhonov regularization".
- Consider how the two penalties differentially treat $\beta = (1, 0)$ and $\beta = (1/\sqrt{2}, 1/\sqrt{2})$.

$$||(1,0)||_2 = 1 = ||(1/\sqrt{2}, 1/\sqrt{2})||_2 = 1$$
$$||(1,0)||_1 = 1 < ||(1/\sqrt{2}, 1/\sqrt{2})||_1 = \sqrt{2}$$

- L2 has no preference between sparse and dense solutions.
- L1 prefers sparse.

Bound-constrained

- Another way to see this is through the equivalent characterization

$$\hat{\beta} = \mathrm{argmin}_\beta \, ||y - X\beta||^2 + \lambda \sum_j c(\beta_j)$$
$$= \mathrm{argmin}_\beta \, ||y - X\beta||^2 \quad \text{subject to} \quad \sum_j c(\beta_j) \leq \eta$$

- where the two approaches can be related using Lagrange multipliers.
- Solutions are characterized by the equilibrium between loss and constraint.
- The shape of the constraint region determines the nature of the solution.

Norm-based bound constraints

- In the case of constraints derived from norms

$$c_q(\beta) = \sum_{j=1}^{d} |\beta_j|^q$$

The shapes of the constraint regions can vary substantially for different q.

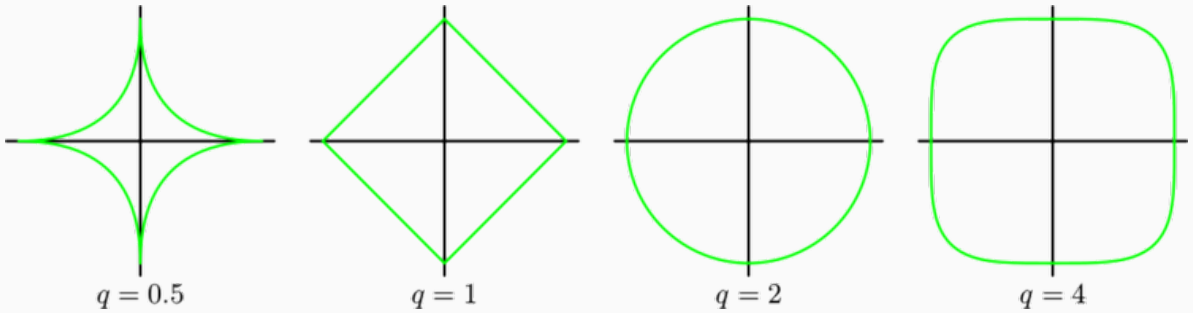

Figure 11: comparing q-norms

An alternative formulation for regularization

- Ridge: for every λ, there is an s such that $\hat{\beta}_\lambda^R$ solves:

$$\text{minimize}_\beta \left\{ \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{i,j} \right)^2 \right\} \text{ subject to } \sum_{j=1}^{p} \beta_j^2 < s$$

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- Ridge: for every λ, there is an s such that $\hat{\beta}_\lambda^R$ solves:

$$\text{minimize}_\beta \left\{ \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{i,j} \right)^2 \right\} \text{ subject to } \sum_{j=1}^{p} \beta_j^2 < s$$

- Lasso: for every λ, there is an s such that $\hat{\beta}_\lambda^L$ solves:

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- Best Subset:

$$\text{minimize}_\beta \left\{ \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{i,j} \right)^2 \right\} \text{ subject to } \sum_{j=1}^{p} \mathbf{1}(\beta_j \neq 0)$$

Visualizing Ridge and Lasso with two predictors

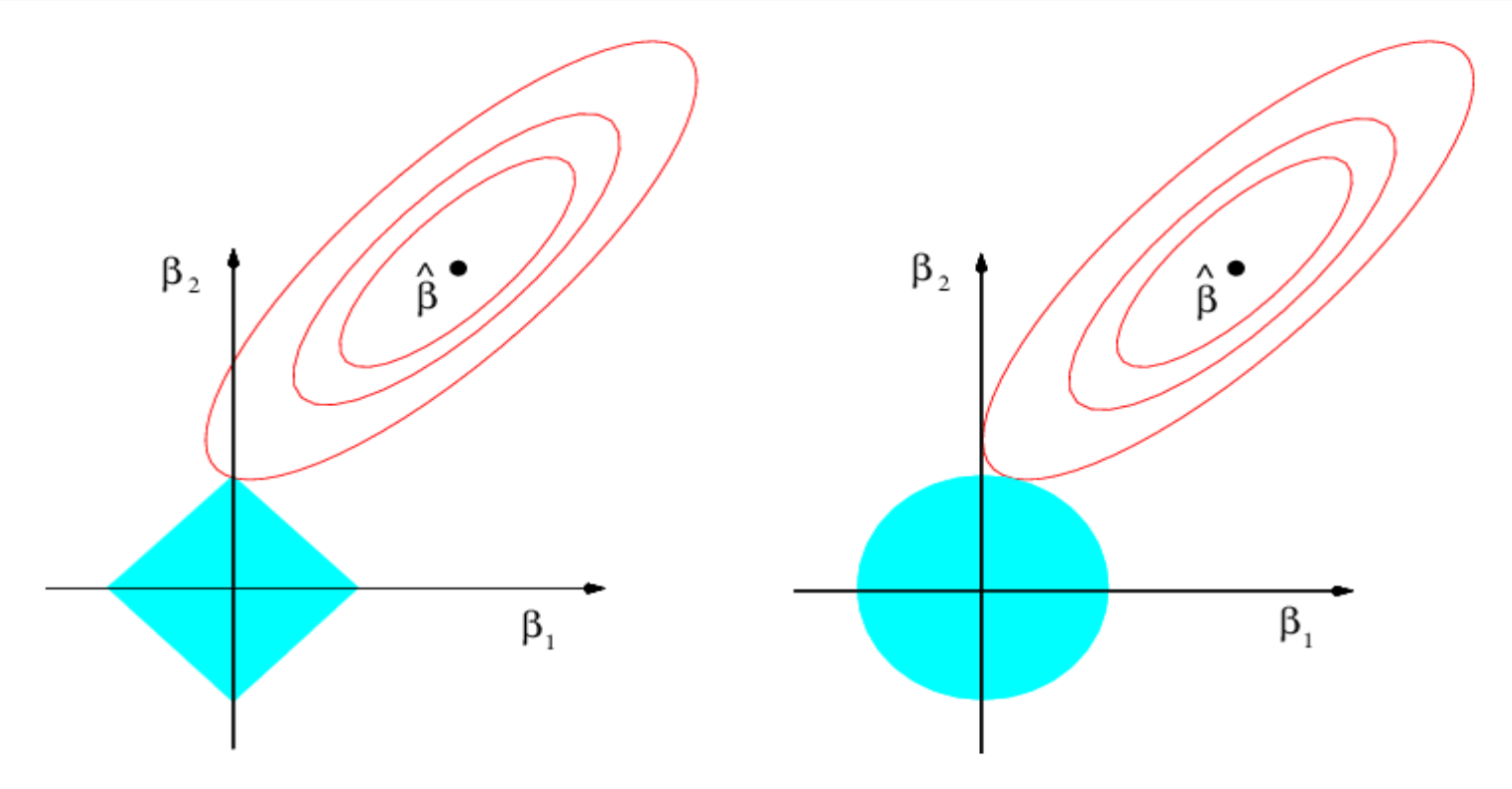

Figure 12: Contours of the error and constraint functions for the lasso (left) and ridge regression (right). The solid blue areas are the constraint regions, $|\beta_1| + |\beta_2| \leq s$ and $\beta_1^2 + \beta_2^2 \leq s$, while the red ellipses are the contours of the RSS.

Intuitive Explanation

- The ellipses that are centered around $\hat{\boldsymbol{\beta}}$ represent regions of constant RSS.

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- However, the lasso constraint has corners at each of the axes, and so the ellipse will often intersect the constraint region at an axis. When this occurs, one of the coefficients will equal zero.
- In higher dimensions, many of the coefficient estimates may equal zero simultaneously

Finding $\hat{\beta}_\lambda$

- Ok, we'll show the formula for $\hat{\beta}_\lambda^{\text{ridge}}$ for now; you'll look at it again on your homework.
- How do we find $\hat{\beta}_\lambda^{\text{lasso}}$?
- Its just an optimization problem, so we could throw it into 'optim'.
- As always, it helps to have derivatives, but unfortunately the constraint is not continuous.
- Dealing with that requires **subgradients**, which we aren't going to bother with.
- Of course, even with a nice solution we still haven't determined λ.
- But at least that's just a scalar, rather than an indicator with 2^p choices.
- To provide some insights, we'll look at a special case now.

A very special case

- Suppose $n = p$ and our matrix of predictors is $X = I$.
- Then, the objective function in Ridge regression can be simplified:

$$\sum_{j=1}^{p}(y_j - \beta_j)^2 + \lambda \sum_{j=1}^{p} \beta_j^2$$

- and we can minimize the terms that involve each β_j separately:

$$(y_j - \beta_j)^2 + \lambda \beta_j^2$$

- It is a non-trivial task to show that:

$$\hat{\beta}_j^R = \frac{y_j}{1 + \lambda}$$

A very special case

- Similar story for the Lasso.
- The objective function in Lasso regression can be simplified:

$$\sum_{j=1}^{p}(y_j - \beta_j)^2 + \lambda \sum_{j=1}^{p}|\beta_j|$$

- and we can minimize the terms that involve each β_j separately:

$$(y_j - \beta_j)^2 + \lambda|\beta_j|$$

- It is trivial to show that:

$$\hat{\beta}_j^L = \begin{cases} y_j - \frac{\lambda}{2} \text{ if } y_j > \lambda/2 \\ y_j + \frac{\lambda}{2} \text{ if } y_j < -\lambda/2 \\ 0 \text{ if } |y_j| < \lambda/2 \end{cases}$$

Different Shrinkage

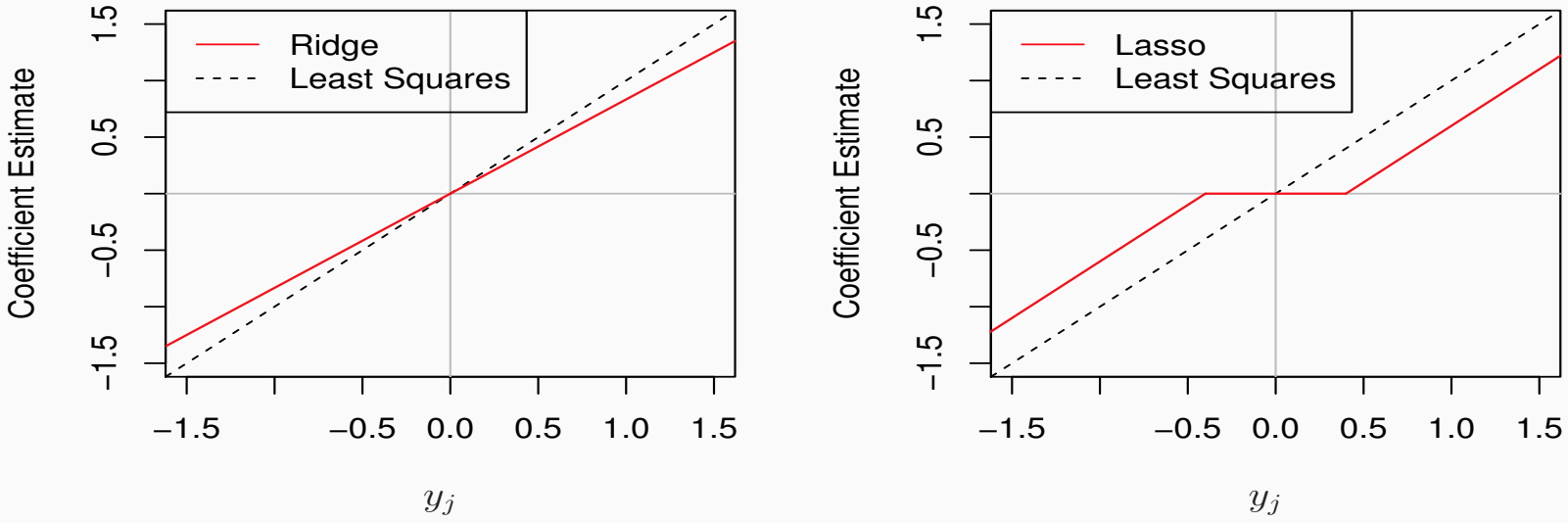


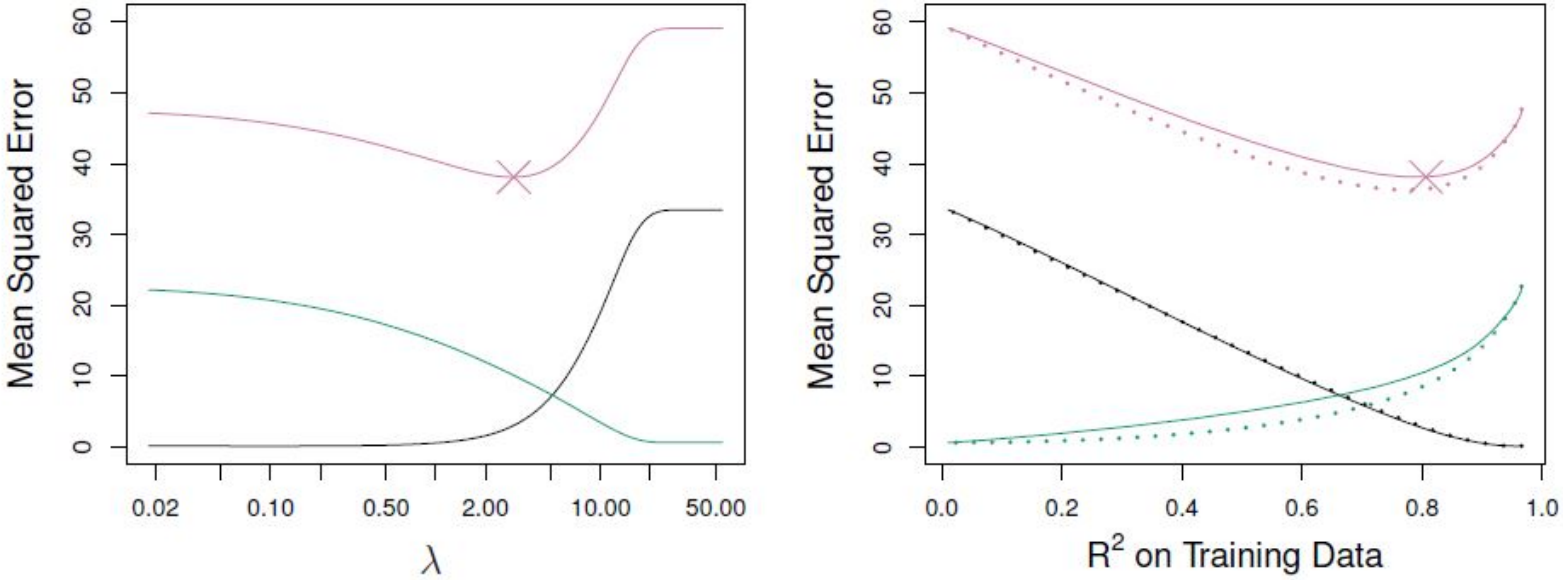
Figure 13: The ridge regression and lasso coefficient estimates for a simple setting with $n = p$ and X a diagonal matrix with 1's on the diagonal. Left: The ridge regression coefficient estimates are shrunken proportionally towards zero, relative to the least squares estimates. Right: The lasso coefficient estimates are soft-thresholded towards zero.

Soft thresholding

- We can see that ridge regression and the lasso perform two very different types of shrinkage.
- In ridge regression, each least squares coefficient estimate is shrunken by the same proportion.
- In contrast, the lasso shrinks each least squares coefficient towards zero by a constant amount, $\lambda/2$; the least squares coefficients that are less than $\lambda/2$ in absolute value are shrunken entirely to zero.
- The type of shrinkage performed by the lasso in this simple setting is known as **soft-thresholding.**

When is the Lasso better than Ridge?

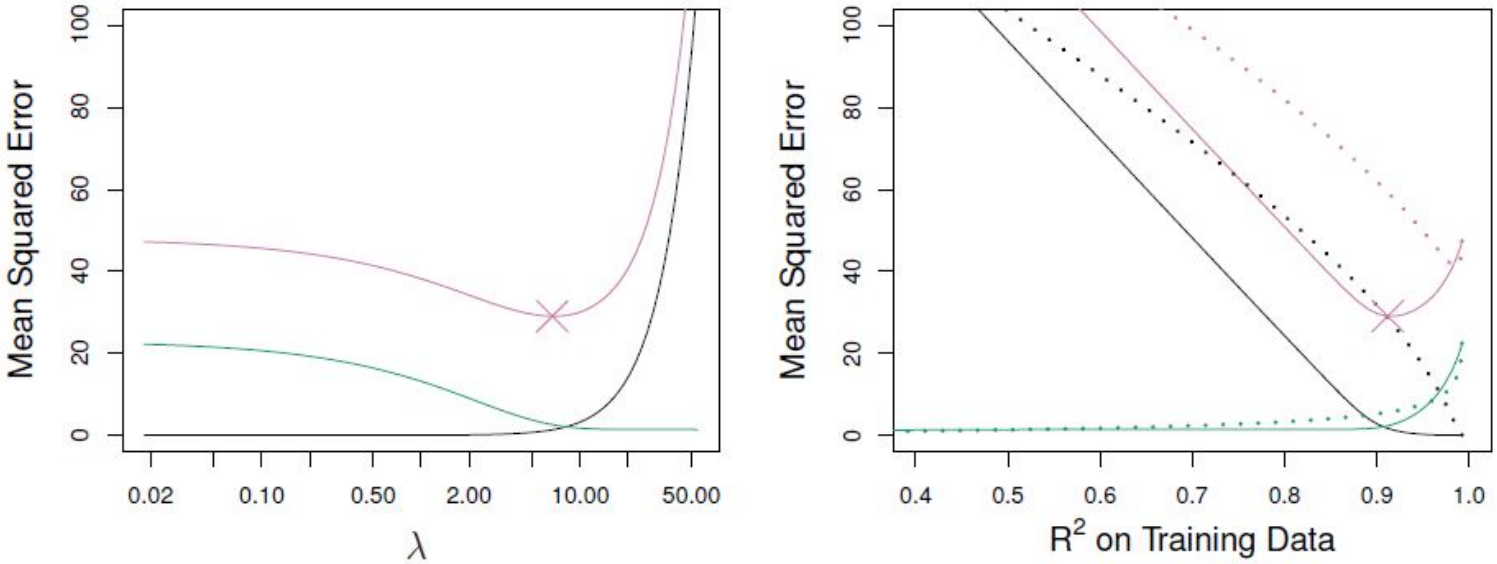
Example 1. All coefficients β_j are non-zero.



- Bias, Variance, MSE. The Lasso (—), Ridge (· · ·).
- The bias is about the same for both methods.
- The variance of Ridge regression is smaller, so is the MSE.
- Rule of thumb: Lasso typically no better than ridge for prediction when most variables are useful for predition.

When is the Lasso better than Ridge?

Example 2. Only 2 of 45 coefficients β_j are non-zero.



- Bias, Variance, MSE. The Lasso (—), Ridge ($\cdots$).
- The bias, variance, and MSE are lower for the Lasso.
- Rule of thumb: Lasso especially effective when most variables are not useful for prediction.

Bayesian Interpretation

- Ridge: $\hat{\beta}_j^R$ is the posterior mode, with a Normal prior on $\boldsymbol{\beta}$.
- Lasso: $\hat{\beta}_j^L$ is the posterior mode, with a Laplace prior on $\boldsymbol{\beta}$.

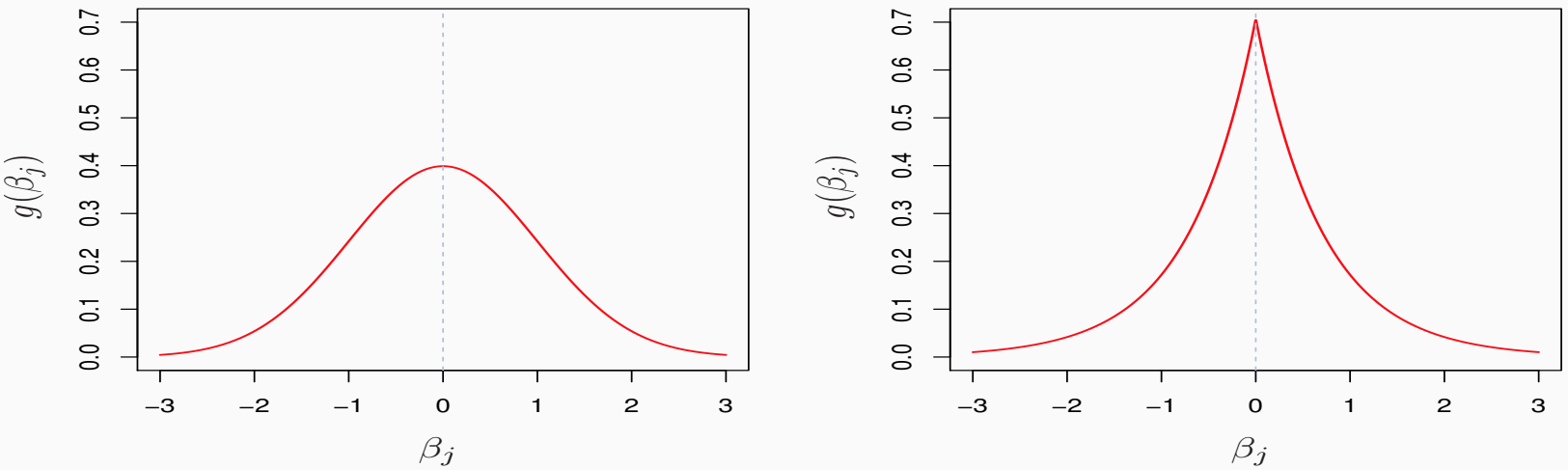


Figure 14: Left: Ridge regression is the posterior mode for $\boldsymbol{\beta}$ under a Gaussian prior. Right: The lasso is the posterior mode for $\boldsymbol{\beta}$ under a double-exponential prior.

Elastic Net

Elastic Net

- Elastic Net ($0 < \alpha < 1$) can outperform Lasso.
- When does this happen?
- An intuitive answer is: when the covariates are grouped.

HIV Drug Resistance Example

- PNAS paper studies in vitro drug resistance of $n = 1057$ HIV-1 isolates to protease and reverse transcriptase mutations.
- Here we focus on Lamivudine (a Nucleoside RT inhibitor). There are $p = 217$ (binary) mutation variables.
- Paper compares 5 different regression methods: decision trees, neural networks, SVM regression, OLS and LASSO.

Leukemia Classification

- Leukemia Data, Golub et al. Science 1999
- There are 38 training samples and 34 test samples with total $p = 7129$ genes.
- Record the expression for sample i and gene j.
- Tumors type: `AML` or `ALL`.
- Golub et al. used a Univariate Ranking method to select relevant genes.

The $p \gg n$ problem and grouped selection

- Microarrays: $p \approx 10,000$ and $n < 100$. A typical "large p, small n"' problem (West et al. 2001).
- For those genes sharing the same biological "pathway", the correlations among them can be high. We think of these genes as forming a group.
- What would an "oracle" do?
- Variable selection should be built into the procedure.
- Grouped selection: automatically include whole groups into the model if one variable amongst them is selected.

Limitations of Lasso

1. If $p > n$, the lasso selects at most n variables!
2. The number of selected **genes** is bounded by the number of **samples**.
3. Grouped variables: the lasso fails to do grouped selection. It tends to select one variable from a group and ignore the others.

Elastic Net Regularization

$$\hat{\boldsymbol{\beta}}^{EN} = \underset{\beta}{\operatorname{argmin}} \, ||\mathbf{y} - \mathbf{X}\boldsymbol{\beta}||_2^2 + \lambda_2 \, ||\boldsymbol{\beta}||^2 + \lambda_1 \, ||\boldsymbol{\beta}||_1$$

- The ℓ_1 part of the penalty generates a sparse model.
- The quadratic part of the penalty
  1. Removes the limitation on the number of selected variables;
  2. Encourages grouping effect;
  3. Stabilizes the ℓ_1 regularization path.

Geometry of Elastic Net

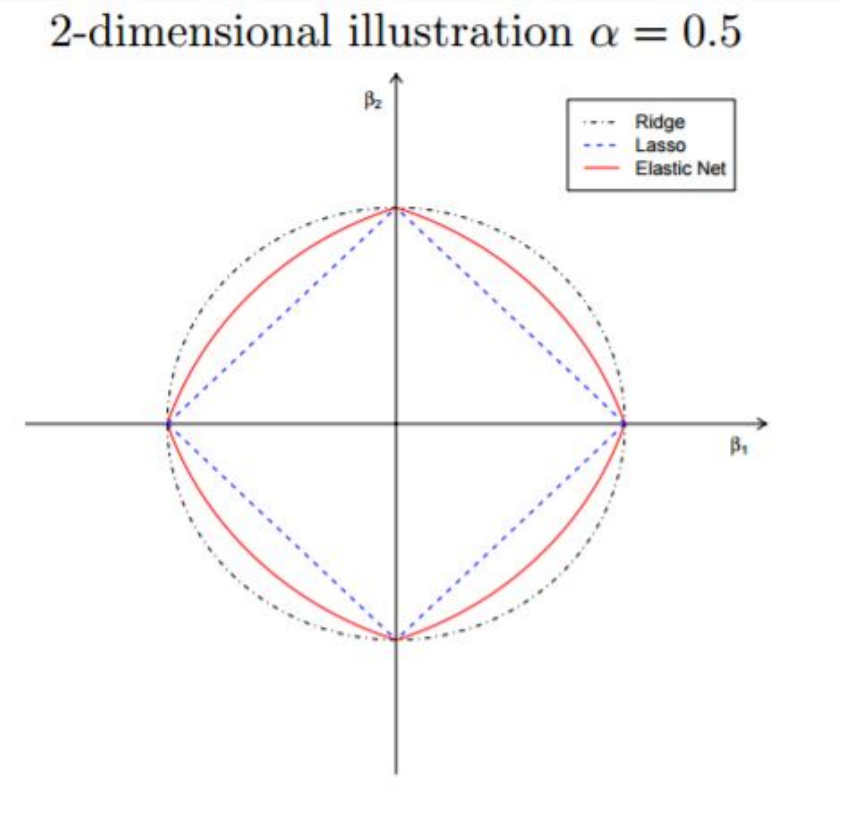


Figure 15

The EN penalty:

$$J(\boldsymbol{\beta}) = \frac{1}{2}(1-\alpha)\,||\boldsymbol{\beta}||_1 + \alpha\,||\boldsymbol{\beta}||^2$$

(where $\alpha = \lambda_2/(\lambda_2 + \lambda_1)$).
Solve:

$$min_{\boldsymbol{\beta}}\,||\mathbf{y} - \mathbf{X}\boldsymbol{\beta}||^2 \text{ s.t. } J(\boldsymbol{\beta}) \leq t$$

- **Singularities** at the vertexes (necessary for sparsity)
- **Strict convex edges.** The strength of convexity varies with α (grouping)

A simple illustration: Elastic Net vs. Lasso

- Two independent "hidden" factors $\mathbf{z}_1$ and $\mathbf{z}_2$.

$$\mathbf{z}_1 \sim U(0, 20), \mathbf{z}_2 \sim U(0, 20)$$

- Generate the response variable $\mathbf{y} = \mathbf{z}_1 + 0.1\mathbf{z}_2 + \mathcal{N}(0, 1)$.
- Suppose we only observe predictors:

$$x_1 = z_1 + \epsilon_1, x_2 = -z_1 + \epsilon_2, x_3 = z_1 + \epsilon_3$$
$$x_4 = z_2 + \epsilon_4, x_5 = -z_2 + \epsilon_5, x_6 = z_2 + \epsilon_6$$

- Fit the model on $\mathbf{x}, \mathbf{y}$.
- An 'oracle' will identify x_1, x_2, x_3 (the $\mathbf{z}_1$ group) as the most important variable.

Let's look at this example in R

.

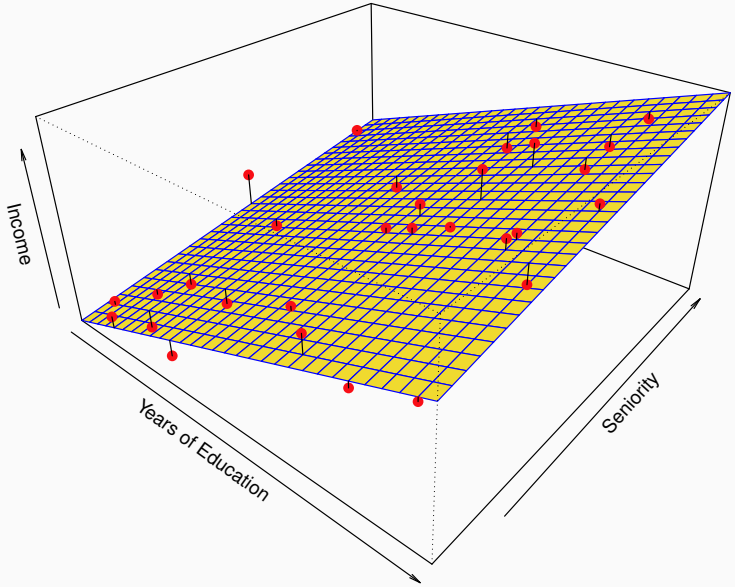
What have we done so far?

Regularization methods: strategies for trading off bias and variance by restricting model flexibility

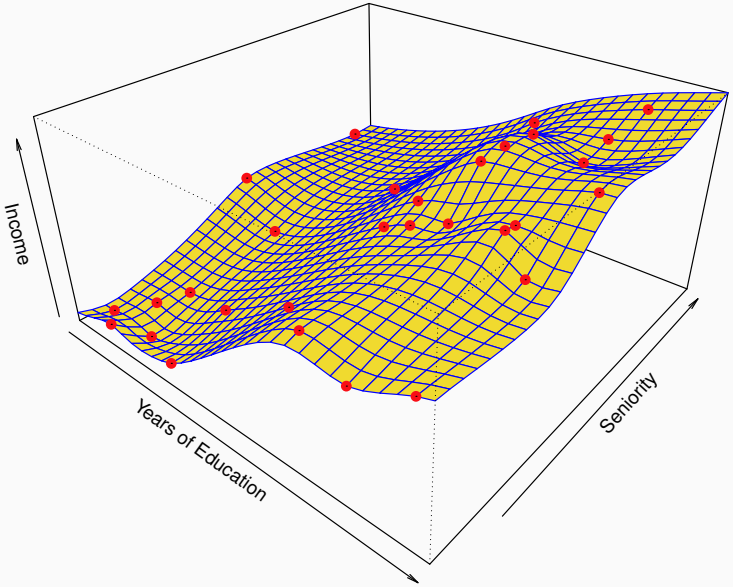
- Variable selection:
  - Best subset selection
  - Forward and backward stepwise selection
- Shrinkage
  - Ridge regression
  - The Lasso (a form of variable selection)
  - Elastic Net (a form of variable selection)
- Dimensionality reduction:
  - **Idea:** Define a small set of M predictors which summarize the information in all p predictors.

Interpretability versus Flexibility

Flexibility vs. Interpretability



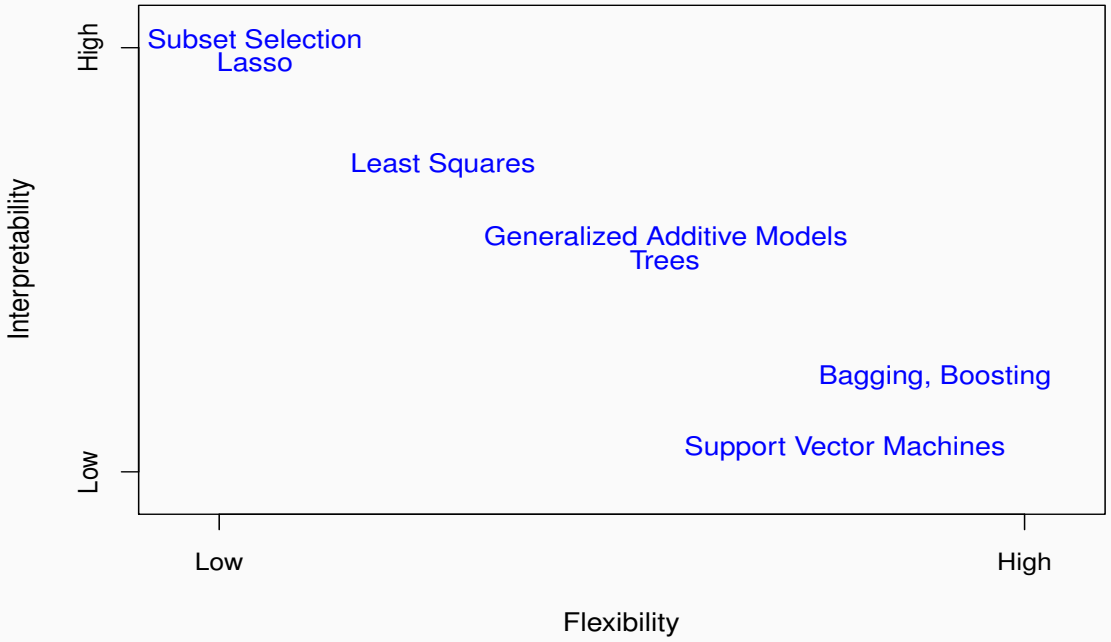
(a) Inflexible but Interpretable: Linear Regression



(b) Flexible but less interpretable: Rough Thin Plate Splines

Prediction Accuracy vs. Model Interpretability

- Of the handful of methods we learn, some are less flexible (more restrictive), some are more flexible.
- For example, linear regression (or LASSO) are more restrictive: allow small variety in $f(x)$.
- Thin-plate splines: More flexible.



Why use a more restrictive method instead of a flexible method?

- If we are mainly interested in **inference**, then restrictive models are much more interpretable.
- Splines / Boosting: difficult to understand how any individual predictor is associated with the response.

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- If we are mainly interested in **inference**, then restrictive models are much more interpretable.
- Splines / Boosting: difficult to understand how any individual predictor is associated with the response.
- LASSO: Many predictors, only a few should be interesting/ relevant. (e.g. Genes / Voxels)
- LASSO: Linear model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \epsilon$ with a penalty for large $||\boldsymbol{\beta}||$.
- LASSO: sets a few β_i's to exactly zero.
- less flexible approach but more interpretable than linear regression,
- Because in the final model y will be a function of a small subset of β_i's: namely, those with nonzero coefficient estimates.

Summary

- Sometimes we are only interested in prediction, and interpretability doesn't matter.
- Develop an algorithm to predict the price of a stock, our sole requirement for the algorithm is that it predict accurately – interpretability is not a concern.
- In this setting, we might expect that it will be best to use the most flexible model available.

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- This is not always the case! People do care about inference / interpretability !!
- Hopefully, you will realize that by the end of this course.
- Throwing the most complicated model at the data is not always the best idea.